Module 04: Cognition in Ms

Learning Objectives

1. Define **Cognition** within the context of Ms.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of Ms to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **Ms** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between [Previous Topic] and [Next Topic].

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In Ms, we see Cognition manifest in: * **Specific Example 1**: [Add domain-specific example here] * **Specific Example 2**: [Add domain-specific example here]

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.